

PIMS_AlgoHCP: Multi-Attribute Entity Resolution, Digital Signatures, and Managerial Verification Architecture

1. Executive Overview & Business Problem Context

In healthcare CRM platforms (such as PIMS), unique identifiers like mandatory government License IDs (PRC License Number) or exact Birthdates are often uncollectible due to privacy regulations and mobile field constraints. Standard deterministic database matching fails because variations in doctor names (e.g., *Dr. Santa Maria Cruz* vs. *Dr. St. Maria Cruz* vs. *Dr. Santa M. Cruz, M.D.*), medical specialties, and hospital titles create false negatives or massive record duplications.

To solve this problem seamlessly, the system implements a **Weighted Multi-Attribute Probabilistic Machine Learning Engine** coupled with a non-linear **Sigmoid Neural Calibration Engine**, interactive **Digital Signature Pads**, and a **Hierarchical Managerial Approval Portal**. This master document provides a comprehensive technical breakdown of how the engine works, mathematical model mechanics, historical fixes implemented, and the mobile app integration blueprint.

2. Mathematical Model Mechanics & Calibrated Sigmoid Curve

The machine learning model operates as a **Probabilistic Supervised Linkage Classifier** calibrated via a non-linear Logistic Sigmoid activation function. Instead of evaluating fields in isolation, the model extracts a high-dimensional feature vector across all input fields, normalizes field importance weights dynamically, and computes a calibrated match probability percentage $P(\text{Match} \mid X)$ between 0.0% and 100.0%.

Mathematical Model Equations:

1. Linear Weighted Feature Score: $S_{raw} = \sum (w_i \cdot s_i)$ where $\sum w_i = 1.0, s_i \in [0, 1]$

2. Logit Transformation (z-score): $z = 6.5 \cdot (S_{raw} - 0.52)$

3. Sigmoid Neural Probability Calibration: $P(\text{Match} \mid X) = 1.0 / (1.0 + e^{-z})$

4. Output Percentage Confidence Score: $Confidence \% = round(P(\text{Match} \mid X) \cdot 100, 1)$

3. Multi-Attribute Field Weight Allocation Matrix

Attribute Field	Weight	Algorithmic Metric	Matching Function
Doctor Full Name	36.4%	Jaro-Winkler + Soundex + Token-Set	Title stripping, 'St./'Santa' mapping
Primary Specialty	18.2%	Token Overlap + Jaro-Winkler	Medical term synonym overlap
Primary Hospital	18.2%	Institutional Jaro-Winkler	Abbreviation mapping ('Hosp' -> 'Hospital')
City / Municipality	9.1%	Jaro-Winkler Distance	Geographic location similarity
Secondary Hospital	4.5%	Token Jaro-Winkler	Secondary clinic alignment
Street Address	4.5%	Normalized Edit Ratio	Barangay/Street text distance
Contact Number	4.5%	Numeric Digit Extractor	Exact & 7-digit suffix match
Email Address	4.5%	Exact / Domain Jaro-Winkler	Handle & domain similarity
Doctor Signature	Mandatory	Base64 Canvas PNG Extractor	Immutable True-Only-One Signature Lock

4. Comprehensive History of System Improvements & Resolved Fixes

The system underwent rigorous architectural refinement to achieve production-grade stability, security, and multi-device usability:

Fix 1: Dynamic Origin API Resolution (Multi-Device Access)

Previously, hardcoded API endpoints ('http://localhost:8080') broke submissions when accessed from external LAN IPs (e.g., 192.168.0.96). Updated client JavaScript (web/app.js) to dynamically resolve `API_BASE = window.location.origin + '/api'`, allowing phones, tablets, and LAN computers to connect seamlessly.

Fix 2: Mandatory Field Validation & Warning Pop-Up

Enforced strict mandatory validation across all 8 doctor fields + Digital Signature. If any field is blank, client-side submission is blocked, empty inputs outline in red, and a prominent **■■ Mandatory Fields Missing Modal** pops up. Backend server.py rejects incomplete submissions with HTTP 400 Bad Request.

Fix 3: 8 Interactive Philippine Demo Presets & Expanded Dataset

Built 8 interactive preset buttons on the UI to test all confidence tiers and string normalization variants: High Match ($\geq 88\%$), 50-50 Match (50-87%), Low Match ($< 50\%$), Honorific Shift (Dr./Dra.), Surname Compound (Dela/De La), Santo/Sto. Shift, Suffix Shift (Jr./Junior), and Delos/De Los Compound Shift.

Fix 4: Unthrottled Real-Time API Engine

Removed API rate-limiting restrictions to allow smooth, uninhibited real-time background duplicate detection as MedReps type rapidly into text fields across multiple mobile devices.

Fix 5: Standalone Microservice Shield & Security Guard (security.py)

Added `hcp_matcher/security.py` providing signed HMAC-SHA256 JWT Bearer tokens and Cython C-extension compilation readiness (`.so/.pyd` bytecode), protecting algorithm weights and master records from unauthorized reverse-engineering or hacking.

Fix 6: New Doctor Canonical Verification Queue & Dictionary Auto-Commit

When a new doctor ($< 50\%$ score) is encoded, the system creates a draft profile (`PENDING_MANAGERIAL_VERIFICATION`) and routes it to the Managerial Portal. Upon manager approval (`VERIFY_AND_LOCK_CANONICAL`), the doctor profile is locked and auto-committed to the **100% Verified Master Dictionary** (`DICT-500X`).

Fix 7: Doctor Digital Signature Pad & Immutable True-Only-One Signature Lock

Integrated an interactive HTML5/Touch Digital Signature Pad. Once verified by a Manager, the doctor's signature is locked as `LOCKED_TRUE_ONLY_ONE`. It becomes permanent and immutable, preventing tampering or unauthorized overwriting by future MedRep submissions.

Fix 8: Socket Rebind Fix (`allow_reuse_address = True`)

Resolved `OSError: [Errno 48] Address already in use` during server restarts by configuring `socketserver.TCPServer.allow_reuse_address = True` in `server.py`.

5. Step-by-Step End-to-End Execution Pipeline

1. **Field Ingestion:** MedRep encodes 8 mandatory fields + draws digital signature on canvas.
2. **Real-Time Scan:** Input change listeners trigger debounced background REST call to `/api/match`.
3. **Text Normalization:** `normalizer.py` strips titles (Dr., M.D.) and expands Philippine abbreviations (St. ↔ Santa, Dela ↔ De La).
4. **Feature Calculation:** Parallel Jaro-Winkler, Soundex 4-character phonetic key, and Token-Set Jaccard ratios computed.
5. **Sigmoid Calibration:** Raw score passed through $z = 6.5 \cdot (S_{raw} - 0.52)$ activation function to produce calibrated % score.

6. **Decision Routing:** High matches ($\geq 88\%$) auto-merge; Medium matches (50-87%) & New Doctors ($< 50\%$) route to Managerial Verification Queue with Immutable Signature Lock.

6. HCP Mobile App Integration Blueprint

The backend microservice directly maps to the Flutter mobile app model (HcpProfileSubmission in lib/models/submission.dart):

- hcpFullName / (firstName, lastName) → Doctor Full Name (36.4% Weight)
- specialties[0].specialtyName → Primary Specialty (18.2% Weight)
- workplaces[0].workplaceName → Primary Hospital (18.2% Weight)
- cityMunicipality / provinceName → City Location (9.1% Weight)
- consentSignature → Base64 Digital Signature PNG (Immutable True-Only-One Lock)